Journal of Non-Crystalline Solids 297 (2002) 263–274

JOURNAL OF NON-CRYSTALLINE SOLIDS

www.elsevier.com/locate/jnoncrysol

High-energy X-ray diffraction study of La co-ordination in lanthanum phosphate glasses

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Received 13 June 2001; received in revised form 17 September 2001

Abstract

A series of $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses with $0.05 \leqslant x \leqslant 0.28$ was characterized by X-ray diffraction. A change of the La–O co-ordination number, N_{LaO}, from 8.2 ± 0.4 ($x \cong 0.13$) to 6.2 ± 0.3 ($x \cong 0.25$) was extracted by Gaussian fitting of the first-neighbor distances. The decrease of N_{LaO} is a consequence of the reduction in the number of terminal oxygen (O_T) atoms available for co-ordination to the La^{3+} ions. With increasing x, isolated LaO_n polyhedra are replaced by LaO_n groups that share O_Ts. These changes in the La environments explain the compositional dependence of glass properties, including mass density. The positions of the first maxima in the X-ray scattering intensities also depend on composition. The position of the main peak shifts from 13.5 to 15.5 nm^{-1} for La_2O_3 additions up to $x \sim 0.11$, is constant for $0.11 \leqslant x \leqslant 0.25$, and increases again when $x > 0.25$. The constant peak position correlates with the compositional range where N_{LaO} decreases and a plateau in the mass density occurs. For glasses with $x > 0.11$ a prepeak occurs at ~ 10.5 nm^{-1}. This prepeak is related to distances between two La ions which are co-ordinated to different O_Ts of a common PO_4 neighboring unit.

PACS: 61.10.Eq; 61.43.Fs

1. Introduction

Phosphate glasses containing trivalent rare earth (RE^{3+}) ions are used for a variety of optical and optoelectronic applications [1] and information about the local co-ordination environment of those ions is important for understanding useful optical properties. For example, the effects of the local co-ordination environments in several oxide glasses have been studied by comparing the linewidths of the Eu^{3+} $^5D_0 \rightarrow {}^7F_0$ transition [2]. The narrowest lines were found for the phosphate glasses, indicating a relaxation of the glass network to form more uniform environments around the modifier Eu^{3+} cations. Most optical glasses possess less than 10 mol% RE_2O_3 to avoid concentration quenching effects that limit useful

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0022-3093/02/$ - see front matter
PII: S0022-3093(01)00936-X

fluorescence lifetimes. However, binary rare earth phosphate glasses can be made with over 25 mol% RE_2O_3 and these glasses possess interesting luminescent and magnetic properties [3].

Stable $(RE_2O_3)_x(P_2O_5)_{1-x}$ glasses are readily obtained in the vicinity of the metaphosphate composition ($x = 0.25$) where the phosphate network consists of Q^2 phosphate tetrahedra that link through two bridging oxygens to form chain-like anions [4]. (Phosphate glass structures are commonly described using the Q^k terminology, where k is the number of bridging oxygens that link neighboring PO_4 tetrahedra [4].) There have been many recent studies of the RE co-ordination environments in these glasses [5–13]. For example, X-ray absorption fine structure (EXAFS) spectroscopic studies of glasses with x varying in a range from 0.19 to 0.26 indicate that the RE—O first-neighbor distances decrease with an increase in the rare earth atomic number [6,9,13]. There is an associated decrease in the RE—O co-ordination numbers, N_{REO}, although the scatter in the data is significant [9]. Further evidence for these 'lanthanide contraction' effects have been provided by X-ray diffraction studies of similar near-metaphosphate compositions [14].

A review of the structures of rare earth phosphate crystals indicates that the rare earth co-ordination environment should depend on the glass composition. Rare earth pentaphosphate crystals (REP_5O_{14}) have ultraphosphate networks based on tetrahedra with two (Q^2) and three (Q^3) bridging oxygens that form interconnected rings of different conformations, depending on the size of the RE^{3+} ion [15]. The co-ordination number of the RE^{3+} ion (N_{REO}) is 8 although the symmetry also depends on the ion size [16,17]. The N_{REO} for REP_5O_{14} is equal to the number (M_{TO}) of terminal oxygen atoms from the phosphate network $[M_{TO} = n(O_T)/n(RE)]$. Terminal oxygens (O_T) include doubly-bonded oxygens associated with the Q^3 tetrahedra and non-bridging oxygens associated with the Q^2 tetrahedra [4]. All O_Ts occupy positions in P—O_T—RE bridges and the REO_n polyhedra do not share any of the common O_T atoms.

There are two crystalline RE-metaphosphate (REP_3O_9) polymorphs [18]. In the orthorhombic form, distorted REO_8 groups link neighboring metaphosphate chain anions. Of the 8 oxygens that co-ordinate each RE^{3+} ion, four are common to neighboring RE-polyhedra, two of them possess longer RE—O bonds. For example, in crystalline NdP_3O_9, two Nd—O distances are larger by about 30 pm than the remaining 6 Nd—O distances [16]. Distorted RE-octahedra form in the lower density monoclinic forms. Polyhedra associated with smaller RE^{3+} ions, e.g., in ErP_3O_9 [19], do not share common oxygens.

The question we ask in the present study is whether or not the differences in the La co-ordination environments indicated in the structures of RE-phosphate crystals exist in glasses with compositions that range from ultraphosphate to metaphosphate stoichiometries. Changes in the modifier co-ordination environments have been detected for phosphate glasses with bivalent metal oxides MeO [20]. Anomalies in the compositional dependencies of properties, including the mass densities, in the Zn- and Mg-ultraphosphate glass systems could be explained by changes in the Zn and Mg co-ordination numbers, N_{ZnO} and N_{MgO}, from 4 to 6 in accordance with the decreasing M_{TO} [21–23]. The decrease in N_{MeO} indicates the transition from the compositional range where the ultraphosphate networks possess 'dangling' P—O_T bonds to the range where neighboring Me sites cluster to share O_Ts for satisfying their co-ordination requirements [20]. Here a 'dangling' P—O_T bond refers to the terminal oxygen on a Q^3 tetrahedron that does not co-ordinate any Me site. One new O_T per Q^2 unit is formed as a result of the network-depolymerization effect of the addition of modifier oxides. Accordingly, the Q^2 groups need charge compensation with modifier cations and, thus, their O_Ts are co-ordinated with Me sites. The attraction of the O_Ts from the Q^3 units to a modifier site and the tendency of the MeO_n polyhedra not to share any O_Ts are the reasons suggested [20] for the observed changes in N_{MeO}.

The diameter and valency of the modifying cation determine the compositional dependence of its co-ordination environment in a phosphate glass. For Zn^{2+} and Mg^{2+}, the possible co-ordination numbers range from 4 to 6. For larger modifier cations, such as Ca^{2+}, clear changes of N_{CaO} are not observed [23], but a minimum packing density exists at the transitional composition [22], where $N_{CaO} = 6$ equals M_{TO}. Thus, the ratio $M_{TO} = n(O_T)/n(Me)$ and the Me—O co-ordination numbers affect the structural details of ultraphosphate glasses. M_{TO} is determined from $v \cdot (y+1)/y$, where v is the valency of Me^{v+} ions and y is given by $n(Me_{2/v}O)/n(P_2O_5)$ [20].

Similar changes in the Me—O co-ordination number of a trivalent ion site were found by ^{27}Al nuclear magnetic resonance (NMR) spectroscopy in $(Al_2O_3)_x(P_2O_5)_{1-x}$ glasses, with x in the vicinity of 0.33 [24]. The decrease of N_{AlO} from 5.3 to 4.8 with increasing Al_2O_3 was equivalent to the change in the number of terminal oxygens, as calculated from $M_{TO} = (1+2x)/x$. In this binary system, the average N_{AlO} decreases with increasing Al_2O_3 content to avoid the formation of Al—O—Al linkages. A recent diffraction study of two $(Nd_2O_3)_x(P_2O_5)_{1-x}$ glasses indicated a small Nd co-ordination decrease, from 6.9 ± 0.3 to 6.6 ± 0.3, as x increases from 0.20 to 0.25 [25]. This co-ordination change also follows trends expected from the decrease in numbers of terminal oxygens per Nd^{3+}.

For $(RE_2O_3)_x(P_2O_5)_{1-x}$ glasses with RE^{3+} co-ordination numbers between 6 and 8, the compositional range for studying the changes of N_{REO} should be about $0.10 < x < 0.25$. The earlier studies of RE-phosphate glasses have been limited to compositions near the metaphosphate stoichiometry, approximately $0.17 \leqslant x \leqslant 0.26$. Ultraphosphate compositions, particularly those with $x < 0.15$, are difficult to prepare because of the volatility of P_2O_5 at the requisite melting temperatures [26]. The present work extends the earlier studies by considering the structures of $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses, with $0 \leqslant x \leqslant 0.28$, prepared using sealed silica ampoules to prevent P_2O_5 loss. The La^{3+} cation, the largest of the lanthanides, was chosen in the present study to characterize these structural effects. X-ray diffraction (XRD) is used to extract the information about N_{LaO}. High-energy radiation from a synchrotron is used to achieve a high resolving power in the pair distribution functions. Moreover, as known from an earlier study of the La metaphosphate glass [8], some low-Q features of the scattering intensities can be used to obtain information about medium-range order.

2. Experimental

2.1. Sample preparation

Lanthanum ultraphosphate glasses, $(La_2O_3)_x$-$(P_2O_5)_{1-x}$ batched with $0.05 \leqslant x \leqslant 0.15$, were prepared from reagent grade La_2O_3 and P_2O_5. The P_2O_5 was first purified by vacuum sublimation. The materials were handled and mixed in a dry box before being sealed in flame-dried silica ampoules. The samples were gradually heated to 500 °C for 4 h, then melted at temperatures between 1000 and 1200 °C depending on the glass compositions. Silica ampoules with wall thickness $\geqslant 2$ mm are required at high melting temperatures (>1100 °C). The samples were held at the melting temperature for about 1 h until a homogeneous liquid formed. After melting, the samples were immediately transferred into the dry box to avoid exposure to the ambient.

Phosphate glasses batched with $x = 0.15$, 0.20 and 0.25 were prepared from reagent grade $NH_4H_2PO_4$ and $LaPO_4$. The raw materials were mixed and transferred into either a silica or a platinum crucible, then calcined at 500 °C for 12 h. (Longer calcining times resulted in less P_2O_5 vaporization during subsequent melting.) The glasses were melted in air at temperatures between 1250 and 1350 °C for half an hour and poured onto a preheated stainless steel plate. The glasses were not annealed.

Glass compositions were examined using both energy dispersive X-ray analysis (EDAX) and inductively coupled plasma atomic emission spectroscopy (ICP–AES). The glass compositions were reproducible to ±3% relative. Glasses prepared in silica ampoules or crucibles contain 2–10 mol% SiO_2. Glass samples with SiO_2 contents greater than ~ 6 mol% were not examined in the present study. The densities and refractive indices of the glasses have been reported elsewhere [27]. Table 1 shows both the batch and final compositions of the lanthanum phosphate glasses examined in this study.

Table 1
Batched and analyzed compositions of the $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses (in mol%)

Glass label	Batched	Analyzed		
	La_2O_3	La_2O_3	P_2O_5	SiO_2
LaP3 [8]	0.25	0.253	0.747	0.00
L10	0.25	0.28	0.72	0.00
L9	0.20	0.233	0.72	0.048
L8	0.15	0.21	0.72	0.063
L7	0.15	0.13	0.84	0.032
L6	0.10	0.11	0.85	0.043
L5	0.075	0.07	0.90	0.028
L2	0.05	0.05	0.95	0.00

Finally, a metaphosphate glass from an earlier study was also examined [8]. This sample, designated LaP3, was prepared in a platinum crucible using crystalline LaP_3O_9. The melt was held at a temperature of 1560 °C for 1 h and then poured into a mold and annealed at 600 °C for 0.5 h. A compositional analysis of the glass revealed a slight excess of La_2O_3 ($x = 0.253$) with respect to the metaphosphate composition. The mass density of this glass was 3.22 ± 0.01 g cm^{-3}.

2.2. Diffraction experiments

The X-ray diffraction experiments were performed on the BW5 wiggler beamline at the DORIS III synchrotron (Hamburg/Germany) where an incident photon energy of 120 keV ($\lambda = 0.01033$ nm) was used. Powdered samples were loaded into silica capillaries (2.0 mm diameter, wall thickness of 10 μm). During the measurements the specimens were positioned in a vacuum vessel to suppress air scattering. Since the scattering angles (2θ) are small, the transmission factors are assumed to be independent of the angle. The angular increment in the step-scan mode was 0.05° over the full 2θ-range from 0.35° to 25°. Experimental details and corrections are described elsewhere [28]. The electronic energy window of the solid-state Ge-detector was chosen to pass the elastic line and the full Compton profile but no fluorescence scattering. The dead-time corrections were made with $\tau = 2.4$ μs. A fraction of 0.91 of the incident photons is polarized horizontally. Corrections were made for background, sample holder scattering, and absorption. The scattering intensities are normalized to the structure-independent scattering functions. These functions are calculated by a polynomial approach [29] of the tabulated atomic data of the elastic scattering factors [30]. The Compton scattering is calculated according to [31]. Finally, the Compton fraction is subtracted and Faber–Ziman structure factors, $S(Q)$, are obtained [32].

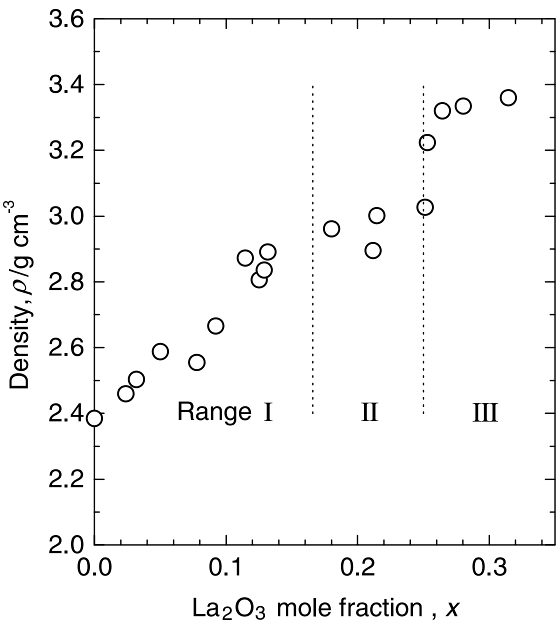

Fig. 1. Compositional dependencies of the mass densities of lanthanum phosphate glasses examined in the present study and reported in [27]. The vertical lines indicate the borders between Range I, Range II, and Range III structures, described in the text. The uncertainty of the measurements is ±0.1 g/cm^3.

3. Results

Fig. 1 shows the mass densities of the $(La_2O_3)_x$-$(P_2O_5)_{1-x}$ glass samples examined in the present study, as well as of samples described in an earlier paper [27]. The initial addition of La_2O_3 to P_2O_5 increases density up to $x \sim 0.12$. Density changes little in the range $0.12 < x < 0.25$ and then increases again when $x > 0.25$.

The X-ray structure factors of the glasses are shown in Fig. 2, along with the $S(Q)$ function of v-P_2O_5 [33]. The $S(Q)$ data change in a systematic way with increasing La_2O_3 content, independent of the residual silica contents of the glasses.

The correlation functions, $T(r)$, are obtained by Fourier transformation (FT) with

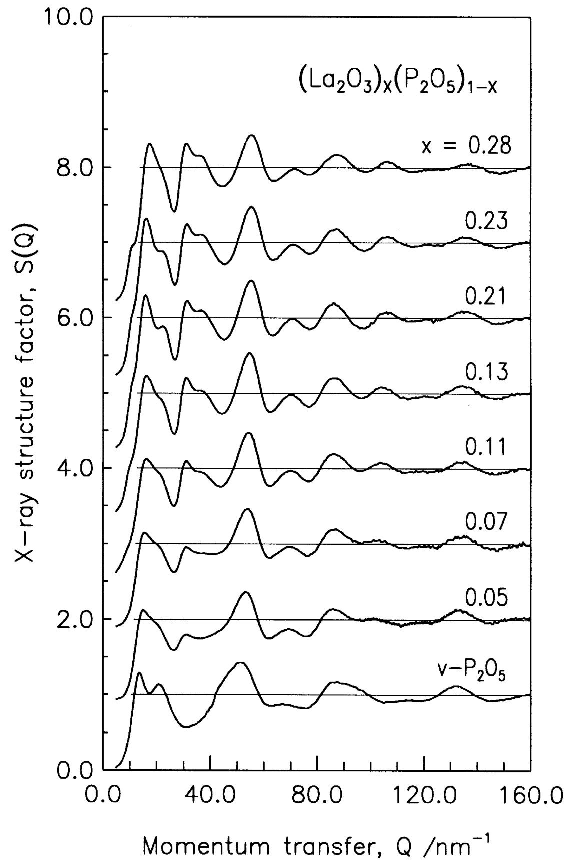

Fig. 2. X-ray structure factors, $S(Q)$, of the $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses and of v-P_2O_5 [33]. The upper curves are shifted for clarity.

$$T(r) = 4\pi r \rho_0 + 2/\pi \int_0^{Q_{\max}} Q \cdot [S(Q) - 1] \times M(Q) \sin(Qr)\, dQ, \quad (1)$$

where ρ_0 is the number density of atoms. $M(Q)$ is the damping function from Lorch [34]: $M(Q) = \sin(\pi Q/Q_{\max})/(\pi Q/Q_{\max})$. The damping reduces the effects of noise and a few spurious features in the data at large scattering angles. The maximum momentum transfer, $Q_{\max}$, was limited to about 240 nm^{-1} $(Q = 4\pi/\lambda \sin\theta)$ due to problems with the background from the sample chamber. However, the synchrotron data are clearly better than those obtained in an earlier study [8] where Ag K_α radiation was used. The XRD measurements of LaP3, the LaP_3O_9 glass $(x = 0.253)$ used in [8], were repeated on the BW5 instrument but not at the same time and also with another wavelength as used for the other samples studied. The peaks in the corresponding $T(r)$ function (cf. Fig. 3) are better resolved than in the older $T(r)$ data [8]. Neutron diffraction data are also available for this sample [8]. The results of the combined analysis

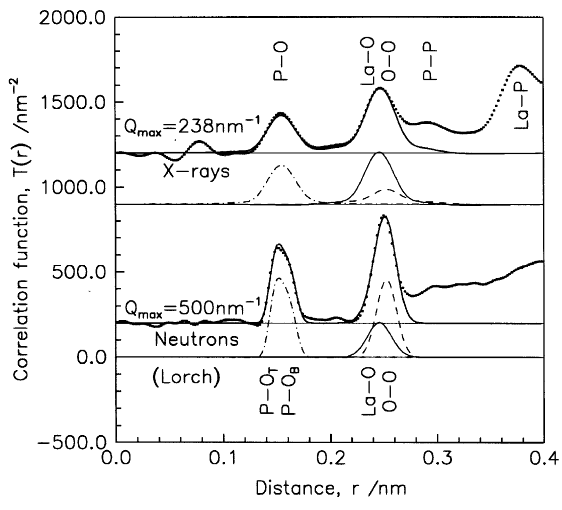

Fig. 3. Comparison of the experimental X-ray and neutron $T(r)$ data (dotted lines) of the LaP3 glass with the model Gaussian fits (solid lines). The partial model functions are shown as peaks already weighted according to the concentration and scattering power and broadened due to the effects of damping and the FT calculation: P—O (dash-dotted lines), La—O (solid lines), and O—O (dashed lines). The upper functions are shifted for clarity.

justify some of the assumptions, described below, used to fit the diffraction data for the present samples.

The structural effects of the residual silica are accounted for in our analyses. We assume that all Si atoms are located in Q^4 Si-tetrahedra, each with 4 bridging oxygens (O_B). These oxygens are bonded either to a second Si or to a P atom but not to any La atom. (This model is consistent with the ^{29}Si NMR spectra collected from several of these glasses which show only Q^4 Si-tetrahedra [35].) Thus, all Si atoms are assumed to have 4 O neighbors and the O atoms which belong to the silica fraction have 6 first-neighbor O atoms. We will see that the structural effects of the residual SiO_2 are small and that alternative model assumptions about how it is incorporated are not necessary.

The parameters of the first P—O, La—O and O—O distances are determined by Gaussian fitting of the correlation functions. In Fig. 4 the resulting model peaks are compared with experimental $T(r)$ data. The effects of damping and truncation by $Q_{\max}$ in the FT procedures are taken into account by the method described in [36]. For fitting the $T(r)$ data, the Marquardt algorithm [37] is used where co-ordination numbers, N_{ij}, mean distances, r_{ij}, and full widths at half maximum (fwhm), Δr_{ij}, are the parameters of the model Gaussian functions. Two Gaussian functions are used for fitting the P—O peak, one representing the terminal oxygens, P—O_T and one representing contributions from the bridging oxygens, P—O_B. (Similar split P—O peaks are found for phosphate glasses, including the LaP_3O_9 glass [8], analyzed by high resolution ($Q_{\max} \cong 500$ nm^{-1}) neutron diffraction experiments [38].) In the fits, the differences of the mean P—O_T and P—O_B bond lengths are fixed to 11 pm. The ratios of the fractions of the P—O bonds are fixed according to the degree of network depolymerization. The total N_{PO} numbers and the mean r_{PO} distances were determined and are given in Table 2. Neither parameter changes with x, in agreement with previous studies of several phosphate systems [21–23,38]. The parameters of the Si—O peak are fixed with $N_{SiO} = 4$, $r_{SiO} = 0.162$ nm [39] and $\Delta r_{SiO} = 0.018$ nm. The corresponding peaks are not shown in Fig. 4. (For the sample with the highest silica contamination (sample L8), such peaks would represent only 5% of the total peak area.)

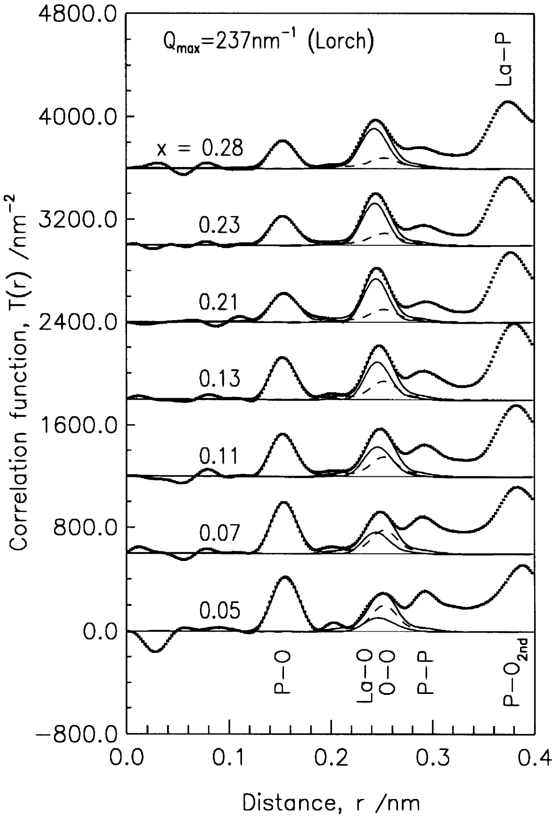

Fig. 4. Experimental X-ray $T(r)$ functions of the La phosphate glasses (dotted lines) in comparison with the model Gaussian fits (solid lines). The partial model functions are shown as peaks already weighted according to concentration and scattering power and broadened due to the effects of damping and FT calculations: La—O (solid lines), and O—O (dashed lines). The upper functions are shifted for clarity.

The La—O first-neighbor peak is decomposed into two Gaussian functions according to our experience with its asymmetry [8]. This does not imply the existence of two La—O distances. Some of the 6 parameters of the two Gaussians functions are fixed: The second distances $^2r_{LaO}$ are equal to 248 pm and their widths $^2\Delta r_{LaO}$ are equal to 30 pm. These values are not given in Table 2. No evidence for a second La—O peak, near 0.27 nm, is apparent. Longer RE—O distances are found in orthorhombic crystals like NdP_3O_9 [16]. Such La—O

Table 2
Results of the fits of the experimental $T(r)$ data

Sample	x	$^*N_{PO}$	$^*r_{PO}$	$^*N_{LaO}$	$^*r_{LaO}$	$^1N_{LaO}$	$^1r_{LaO}$	$^1\Delta r_{LaO}$	$^PN_{OO}$	r_{OO}	Δr_{OO}	$^{Si}N_{OO}$
LaP3	0.253	4.02(15)	154.9(10)	6.2(3)	246(2)	5.2(2)	245.5(8)	24(3)	3.91(10)	252.5(8)	20(3)	–
L10	0.28	4.05(15)	153.1(10)	6.1(3)	244(2)	5.3(2)	242.0(10)	22(3)	3.89**	252*	20*	–
L9	0.23	3.90(15)	153.1(10)	6.4(3)	243(2)	5.2(2)	243.0(10)	20(2)	3.94**	252*	20*	0.07**
L8	0.21	3.80(15)	153.8(10)	6.8(3)	245(2)	5.4(3)	244.0(15)	19(2)	3.95**	252*	20*	0.10**
L7	0.13	3.70(10)	154.0(10)	8.2(4)	247(3)	5.7(3)	244.0(20)	19(3)	4.34**	252*	20*	0.04**
L6	0.11	3.50(10)	153.7(10)	7.9(5)	247(4)	5.7(4)	244.0(30)	23(4)	4.38**	252*	20*	0.05**
L5	0.07	3.62(10)	154.6(10)	7.7(10)	246(5)	5.5(8)	242.0(50)	21(5)	4.53**	252*	20*	0.04**
L2	0.05	3.70(10)	155.3(10)	7.5(20)	249(8)	4.0(10)	242.0(80)	23(8)	4.65**	252*	20*	–

The distances (r_{ij}) and full widths at half maximum (Δr_{ij}) of the Gaussian functions are given in pm. The parameters $^*N_{ij}$ and $^*r_{ij}$ denote total co-ordination numbers and mean distances where, for shortness, only the parameters of the first La—O peak are given ($^1N_{LaO}$, $^1r_{LaO}$, $^1\Delta r_{LaO}$). Numbers marked by asterisks are fixed during the fits, those with ** are calculated (cf. the text). For the sample LaP3 [8] the fit is performed using neutron and XRD data. In all other cases only the XRD data are available.
** The O—O co-ordination numbers of the PO_4 and SiO_4 tetrahedra are calculated separately and weighted according to the $(La_2O_3)_x(P_2O_5)_{1-x}$ and SiO_2 fractions (cf. Table 1), respectively.

distances cannot be extracted from the total $T(r)$ data due to overlapping peaks in this r-range. The co-ordination numbers N_{LaO} show a decrease from ~ 8 at $0.05 \leqslant x \leqslant 0.13$ to ~ 6.2 for $x \geqslant 0.25$. The accompanying decrease of the mean La—O distances is rather small, from ~ 247 to ~ 244 pm. The distances r_{PO}, r_{LaO} and r_{OO} for the LaP3 glass are slightly different from those of the similar sample L10 but the corresponding co-ordination numbers are similar.

A fifth Gaussian function approximates the O—O peak. Due to the different contrast, the combined fit of neutron and XRD data of the LaP3 glass allows the resolution of the La—O and O—O contributions (Fig. 3 and Table 2). For the other glasses, the area of the O—O peak is calculated. This peak describes the lengths of the edges of the PO_4 (and SiO_4) tetrahedra. Since each O_B has 6 O neighbors and each O_T has three O neighbors, the average O—O co-ordination number depends on the degree of network depolymerization according to $N_{OO} = 24/(5+y)$ [21], where y is the ratio $n(La_{2/3}O)/n(P_2O_5), y = 3x/(1-x)$. An N_{OO} equal to 6 is used to account for the residual silica. The distances r_{OO} and widths Δr_{OO} are fixed to 0.252 and 0.020 nm, respectively, for the phosphate fraction (cf. with the data from LaP3) and to 0.262 and 0.024 nm, respectively, for the silica fraction [39]. The N_{OO}s given in Table 2 are reduced according to the fractions of the oxygen atoms belonging either to the phosphate or to the silica fraction. The comparison of both N_{OO}s in Table 2 demonstrates that the effects of the silica contamination on the results of the Gaussian fitting are rather small. The combined fit of the neutron and X-ray data for the LaP3 sample gives an N_{OO} of 3.91 which is in good agreement with the expected N_{OO} of 3.99 calculated according to $y \cong 1.016$. The model peaks which correspond with the La—O and O—O distances are shown in Figs. 3 and 4, as well. Except for the samples L5 and L2 the La—O distances dominate the peak at 0.250 nm. Thus, for the other samples a determination of La—O co-ordination numbers and distances is well justified and the change observed for N_{LaO} is a real effect even though assumptions about the O—O distances had to be made. Uncertainties associated with the assumptions about the SiO_2 fraction can have only minor effects. The Si—O and O—O peaks which belong to the SiO_4 tetrahedra are not shown because they do not increase much above the base lines.

4. Discussion

The effects of glass composition on the La—O co-ordination environments (Table 2) can be explained using the concepts developed by Hoppe et al. [20–23] for other ultraphosphate glasses as reviewed in Section 1. The apparent decrease in La—O co-ordination numbers, N_{LaO}, is related to the decrease in the number of terminal oxygens, M_{TO}, available for co-ordination of each La^{3+} site. For glasses with the stoichiometry $(Me_{2/v}O)_x$-$(P_2O_5)_{1-x}$, where Me is a metal modifier and v is the valency of the modifier ion, the number of terminal oxygens per modifying cation is $M_{TO} = v(1/x)$. This equation is re-written for $(RE_2O_3)_x(P_2O_5)_{1-x}$ glasses to determine the dependence of the ratio of terminal oxygens per RE^{3+} cation on the mole fraction (x) of RE_2O_3:

$$n(O_T)/n(RE^{3+}) = (1 + 2x)/x. \quad (2)$$

For isolated RE-polyhedra that bond only to terminal oxygens, the maximum possible RE—O co-ordination number will equal $n(O_T)/n(RE^{3+})$.

Fig. 5 shows the compositional dependence of the La—O co-ordination numbers determined by XRD (circles) and predicted by Eq. (2) (heavy solid line). The vertical dashed lines indicate the compositional range over which N_{LaO} depends on composition according to Eq. (2); this compositional range is identified as 'Range II'. Range I includes glass compositions with $x < 0.166$, where N_{LaO} is independent of x and equal to ~8. Range III includes glass compositions with $x > 0.25$, where N_{LaO} is again independent of x and equal to ~6.2. (A previous study of RE metaphosphate

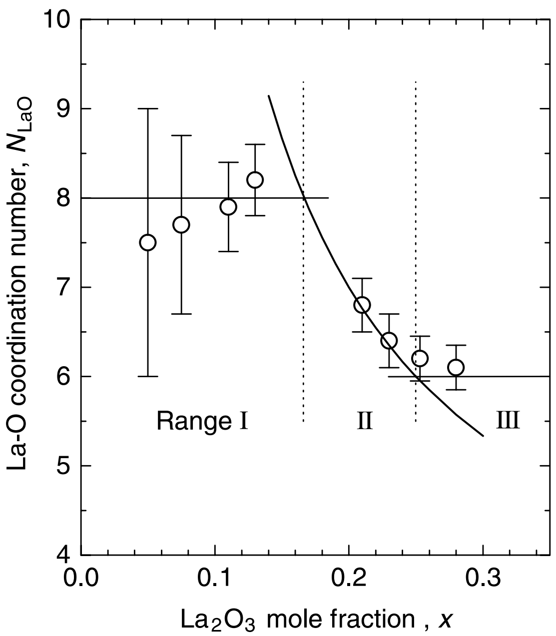

Fig. 5. The effect of composition on the La—O co-ordination numbers (circles) for the $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses. The number of terminal oxygens per La^{3+} (Eq. (2)) is shown as the heavy line. The vertical lines indicate the borders between Range I, Range II, and Range III structures, described in the text.

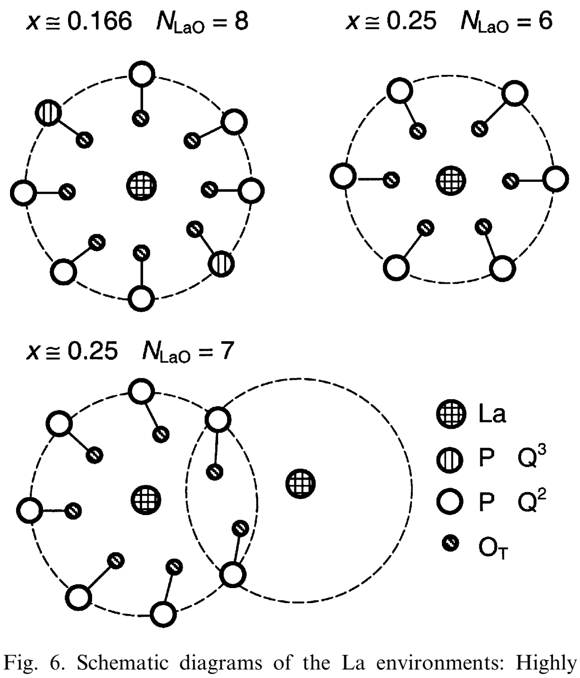

Fig. 6. Schematic diagrams of the La environments: Highly symmetric La co-ordination spheres will form with N_{LaO} of 8 at $x = 0.166$ (upper left) and 6 at $x = 0.25$ (upper right) when all O_Ts are in La—O_T—P bridging positions. The symmetry is reduced if N_{LaO} exceeds M_{TO} and the LaO_n polyhedra tend to cluster through common O_Ts (below).

glasses also indicated that N_{REO} is ~ 6.5 for larger RE^{3+} ions, not the $M_{TO} = 6$ expected for $x = 0.25$ [9].) The same compositional ranges are indicated in Fig. 1 where it is shown that breaks in the compositional dependence of density occur at the transitions between Range I and Range II and between Range II and Range III. Fig. 6 illustrates the La-environments which are favored for $x = 0.166$ and 0.25 with N_{LaO}s of 8 and 6, respectively, and which are also found in RE phosphate crystals [15–17,19]. For glasses with $x \geqslant 0.25$, the N_{LaO} obtained from diffraction experiments is slightly greater than 6 and so a fraction of La-environments with seven oxygen neighbors must exist. Because $N_{LaO} > M_{TO}$ also for $x = 0.25$, these larger La sites must share some O_Ts (cf. Fig. 6 bottom).

Co-ordination numbers obtained from diffraction experiments are mean numbers. We expect that distributions of LaO_n polyhedra, with n between 6 and 9, are present in our glasses. Different La—O distances are characteristic for different LaO_n polyhedra, but such details are not resolved in the experimental La—O correlations. The La—O peaks are characterized by insignificant asymmetry effects which are not further analyzed. Using the reported co-ordination dependence of La^{3+} radii in crystalline compounds [40–42], the La—O bond distances should increase by 13 pm (from 238 to 251 pm) when N_{LaO} is changed from 6 to 8. The difference of the Nd—O distances in the crystals NdP_3O_9 and NdP_5O_{14} [16] is 9 pm. However, the changes of the mean r_{LaO} (or $^1r_{LaO}$) obtained for our glasses are smaller (3 pm) and the mean observed distance of 245 pm (Table 2) is typical for an N_{LaO} of about seven.

There is a loss in polyhedral symmetry when neighboring LaO_n polyhedra share terminal oxygens to form La-clusters (cf. Fig. 6 bottom). Shared O_Ts in La-polyhedral clusters are linked to one P and two La neighbors which makes relaxation of the structure to optimal La—O distances more difficult. Replacing a PO_4 tetrahedron with a LaO_n polyhedron as a next-nearest neighbor group reduces the spherical symmetry of the countercharge around the La^{3+} cation. The decrease of N_{LaO} in Range II with increasing La_2O_3 content avoids the formation of significant concentrations of these asymmetric, higher energy La-clusters. (The lower limit of N_{LaO} is about 6.2. Since M_{TO} equals 6 at the metaphosphate stoichiometry, some O_T sharing must occur and so some LaO_n clusters must be present in these glasses. Smaller RE^{3+} ions have smaller N_{REO} numbers [9] and so might avoid clustering up to metaphosphate compositions.)

The structures of glasses with La_2O_3 contents under about 16 mol% have LaO_n polyhedra with $N_{LaO} \sim 8$. Terminal oxygens on Q^3 phosphate tetrahedra (tetrahedra that possess three bridging oxygens) are expected to participate in the LaO_8 co-ordination environments, as they do in lanthanide ultraphosphate crystals [16,17] (cf. Fig. 6 top). ($^{23}Na \Rightarrow {}^{31}P$ cross-polarization NMR experiments reveal correlations between Na^+ cations and Q^3 tetrahedra in Na ultraphosphate glasses [43], supporting this structural description.) To maintain spherical symmetry around the La^{3+} ions, charge is redistributed among the neighboring terminal oxygens associated with Q^3 and Q^2 tetrahedra, modifying the bond lengths and bond angles of the participating P—O_T and P—O_B bonds [38]. This charge transfer reduces the π-character of the P—O_T bond on a participating Q^3 tetrahedron, also helping to stabilize the ultraphosphate structure [44].

Fig. 5 shows that N_{LaO} remains ~ 8 with decreasing La_2O_3 contents below ~ 16 mol%. This means that increasing numbers of P=O bonds (as x decreases) associated with Q^3 tetrahedra do not participate in the co-ordination environments of the LaO_8 polyhedra. In addition, decreasing La_2O_3 content can lead to edge-sharing between La sites and Q^2 tetrahedra as proposed for other ultraphosphate glass systems (e.g., [20] and references therein). Q^2 units are found close to the La sites in order to maintain charge neutrality. Further reduction in the number of La sites and Q^2 tetrahedra causes further increase of La—La distances. Finally, this leads to the change of Q^2 environments which is illustrated in Fig. 7(a) and (b). The creation of 'dangling' terminal oxygens and the change to edge-shared La–Q^2 linkages are indications for the Range I structures and have been characterized in alkali and alkaline earth ultraphosphate glass systems by a variety of

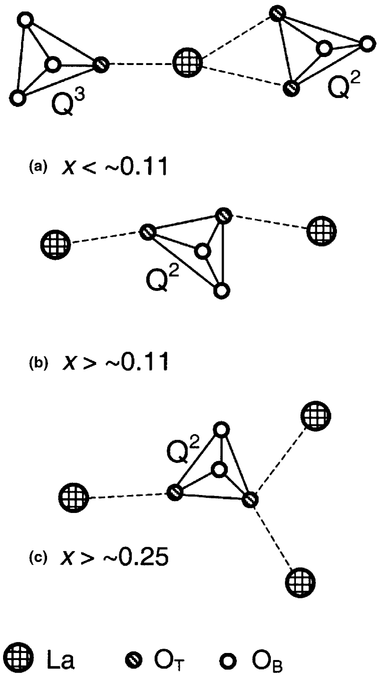

Fig. 7. The compositional dependence of proposed linkages between La ions and phosphate tetrahedra: (a) La—Q^2 edge-sharing linkages, (b) two La sites linked through different O_T corners on a Q^2 tetrahedron with La–La distances of ~ 0.64 nm and (c) two La polyhedra which share a common O_T neighbor with La–La distances of ~ 0.46 nm.

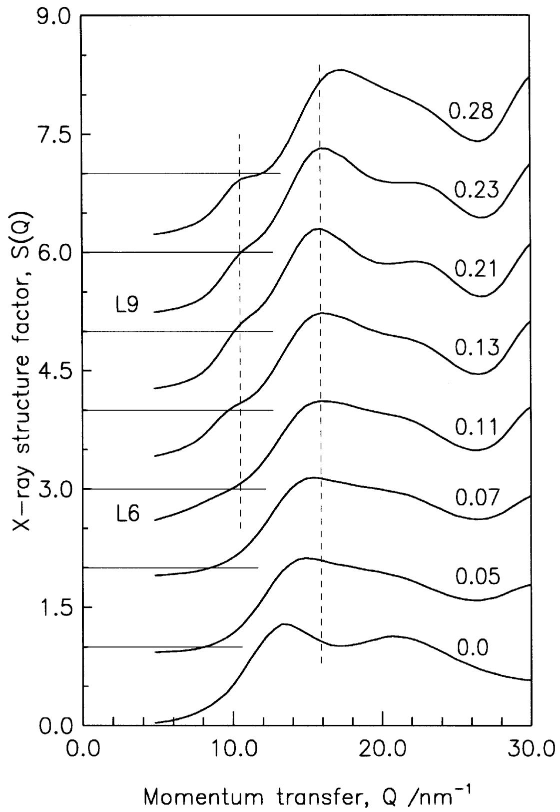

Fig. 8. Expanded X-ray structure factors, $S(Q)$, for the $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses and for v-P_2O_5 [33] in the Q-range for the first diffraction peaks. The vertical lines at $Q = 10.5$ and $15.5\ nm^{-1}$ are guides to the eye. The upper curves are shifted for clarity.

spectroscopic techniques [45–49]. The minima of T_gs reported for alkali ultraphosphate glasses have been explained by corresponding effects of network restructuring in Range I [4,46].

The transitions in structure discussed above are consistent with low-Q features of the diffraction data. The X-ray structure factor, $S(Q)$, of v-P_2O_5 has a double peak as the first maximum. In several MeO—P_2O_5 glasses, this double peak changes to a single first peak at a fraction of 20 mol% MeO [49,50]. This effect is interpreted as a signature for the network restructuring in Range I. Similar observations were also made for Na phosphate glasses by neutron diffraction [51]. The first diffraction peaks of the structure factors for the $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses show a somewhat different behavior (cf. Fig. 8). The first maximum (between about 12 and 24 nm^{-1}) for each La phosphate glass retains the double-peak character of v-P_2O_5. A prepeak, centered near 10 nm^{-1}, is evident in the data from glasses with La_2O_3 contents greater than $\sim$11 mol%.

Other features of the first peaks in Fig. 6 are similar to those reported for MeO—P_2O_5 glasses [50]. The peak at 13.5 nm^{-1} for v-P_2O_5 shifts to 15.5 nm^{-1} in the $\sim$11 mol% La_2O_3 sample. For samples with La_2O_3 contents in the range $0.11 \leqslant x \leqslant 0.25$, the peak position is unchanged. When $x > 0.25$, the peak again shifts to a greater value of Q. The structure factor, thus, exhibits a similar dependence on composition as was found for glass density (Fig. 1) and N_{LaO} (Fig. 5). The break in the shift of the main peak at $x \sim 0.11$ (Range I) indicates restructuring from a network that accommodates edge-shared La–Q^2 linkages to a more compact network (and greater glass density) with corner-linked La–Q^2 linkages. When $x > 0.16$ (Range II), N_{LaO} starts to decrease to avoid La-clustering which results in less dense packing (constant glass density) and a constant position of the main peak. Above $x \sim 0.25$ (Range III), density again increases and the structure factor peak again shifts to greater Q due to structural compaction associated with La—La clusters. The N_{LaO} of ~6.2 of the LaP3 glass indicates that a minor effect of clustering is already present at metaphosphate stoichiometry.

The prepeak which appears at $\cong 10.5$ nm^{-1} (cf. Fig. 6) can be explained from the results of the reverse Monte Carlo (RMC) simulations of the LaP_3O_9 glass structure [8]. The largest contribution to the prepeak comes from a strong first peak in the partial S_{LaLa} (Q) function. This peak corresponds to an La—La distance of ~ 0.64 nm and is attributed to two La sites linked with two different O_Ts on the same PO_4 unit (cf. Fig. 7(b)). Such La—La correlations are expected to form when sufficient La sites and Q^2 units are available to replace the edge-shared La–Q^2 groups in glasses with $x < 0.11$. (The $S(Q)$ data of the $x \sim 0.11$ sample shows a small indication for a prepeak at about 9 nm^{-1}. Barium ultraphosphate glasses with low BaO contents show similar lower Q prepeak positions in their $S(Q)$ functions [50]. It is possible that even greater La—La correlations are present in glasses with the lowest La_2O_3 contents.) The lengths of La—La distances through shared O_Ts in LaO_n clusters in glasses of $x \geqslant \sim 0.25$ (cf. Figs. 6 bottom and 7(c)) are shorter, ~ 0.46 nm. Such La—La pairs do not contribute to the prepeak and their concentrations are low compared with the longer La—La pairs (~ 0.64 nm) illustrated in Fig. 7(b).

5. Conclusions

A decrease of the La—O co-ordination number, N_{LaO}, from $8.2 \pm 0.4 (x \cong 0.13)$ to $6.2 \pm 0.3 (x \cong 0.25)$ in $(La_2O_3)_x(P_2O_5)_{1-x}$ glasses was found by X-ray diffraction experiments. This decrease in N_{LaO} is due to the transition from ultraphosphate glass structures with an excess of terminal oxygen (O_T) atoms available for co-ordination of isolated La^{3+} polyhedra to metaphosphate structures where the LaO_n polyhedra start to share neighboring O_Ts. Structures with all O_T sites in P—O—La bridges are maintained in the transitional range. The decrease of N_{LaO} accounts for the plateau in the compositional dependence of the mass densities. The positions of the main first maxima in the structure factors are constant over a similar compositional range, $0.11 \leqslant x \leqslant 0.25$. The increase in the position of the $S(Q)$ maximum at $x > 0.25$ indicates the formation of a compact structure based on short La—La distances due to shared O_Ts with more efficiently packed La-polyhedra. The change at $x \cong 0.11$ is an indication of network restructuring seen in several earlier studies of alkali and alkaline earth phosphate glasses.

Acknowledgements

The participants from the University of Missouri-Rolla acknowledge the donors of The Petroleum Research Fund, administered by the American Chemical Society, for support of this research.

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